

FR250SDG Rear Helper Spring Kit Installation Manual

Model: FR250SDG (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are met while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (HCS) (3/8" x 1" UNF, Grade 5, Zinc Plated)	2
2	Split Lock Washer (3/8", Grade 5, Zinc Plated)	4
3	Flat Washer (USS) (3/8" x 1", Grade 2, Zinc Plated)	2
4	Weldment Bracket (Painted)	2
5	Hollow Rubber Spring	2
6	Hex Cap Screw (HCS) (3/8" x 1-1/2" UNF, Grade 5, Zinc Plated)	2
7	Nylock Locknut (3/8" UNF, Grade 5, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lift or jack, ensuring the axle is allowed to hang freely. Secure the chassis adequately before proceeding with installation.
- 2. Remove OEM Components:** Remove the vehicle's existing original equipment manufacturer (OEM) bump stop and its associated bushing.
- 3. Assemble Helper Spring Unit:** Assemble the hollow rubber spring with the provided bracket.
- 4. Install Helper Spring Assembly:** Mount the assembled helper spring unit to the vehicle frame, utilizing existing mounting holes and the fasteners supplied in this kit.
- 5. Lower Vehicle:** Carefully lower the vehicle to the ground.

Installation Notes:

- Check for potential interference between the helper spring and any brake lines or cables.
- Utilize Nylock Locknut #7 with the bracket's offset holes as necessary to precisely center the helper spring over the axle.
- If the vehicle is equipped with an OEM peg-style casting, the hollow rubber spring is engineered to make contact with this component. Do not remove the OEM peg.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap:** For light-duty trucks and vans, an ideal clearance of 1/2 inch to 1-1/2 inches should be maintained between the hollow rubber spring and its contact point when the vehicle is unladen.
- **Load Effects:** The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or potentially eliminate this initial clearance.

- Heavy-Duty Applications: For medium-duty and heavy-duty trucks, trailers, and motorhomes, the helper spring is designed to be in constant contact with no discernible gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating.
- Inspect for potential interference between the hollow rubber spring and other vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.